

KAPTURE FINANCE — VOICEBOT HIGH-LEVEL DESIGN (HLD)

Agent Name: Maya | **Target Customer:** Rahul Sharma (Overdue EMI: ₹8,499)

1. CORE ARCHITECTURE & LATENCY BUDGET

To make the AI voice sound completely natural without awkward pauses, the pipeline limits response times to under 800 milliseconds total per turn:

- **Telephony Link:** Handles the raw telephone call routing (Budget: 100ms).
- **Speech-to-Text (STT):** Uses Deepgram Nova-2 to listen to the user and convert their spoken words into text instantly (Budget: 150ms).
- **AI Orchestrator (LLM):** Uses OpenAI GPT-4o as the 'brain' to read the text and decide the next response (Budget: 350ms).
- **Text-to-Speech (TTS):** Uses Cartesia Sonic to convert the text response back into a natural voice with an Indian accent (Budget: 100ms).
- **Database Storage:** A central database holds customer account details, updated live via secure digital links (FastAPI webhooks).

2. STEPS OF THE CONVERSATION (STATE MACHINE)

The voicebot is strictly locked into a step-by-step sequence. It cannot skip steps or reveal private information early:

- **[STATE_DISCLOSE]** -> Initial State. The bot introduces itself ("Maya from Kapture Finance") and states the call is recorded. It identifies the target customer ("Rahul Sharma"). **Compliance Lock:** The bot is completely banned from mentioning money, loans, or debt in this step.
- **[STATE_AUTH]** -> The bot asks the caller to confirm their Date of Birth (DOB) to verify their identity. **Transition Rule:** Unlocks STATE_NEGOTIATE only if the 'verify_customer' API returns a success message. If the caller asks "Why are you calling?", the bot replies: "I can share all account details immediately after verifying your identity for data privacy purposes."
- **[STATE_NEGOTIATE]** -> Unlocked only after successful verification. The bot states the exact overdue amount (₹8,499, 12 days past due) and asks for a payment commitment date.

- **[STATE_CLOSING]** -> The bot notes the agreed payment date, triggers a payment link text message, and concludes the call.
- **[STATE_ESCALATE]** -> If the verification fails twice, or the customer asks for a human, the bot transfers the call immediately.

3. CUSTOMER INTENTS & DATA EXTRACTION

The AI listens to the customer's phrases and automatically categorizes them into one of these buckets:

- **'Will-Pay'**: Customer agrees to clear the balance. The bot extracts the exact promised date.
- **'Cannot-Pay' (Hardship)**: Customer has financial difficulties. The bot offers empathy and notes a future fallback date.
- **'Already-Paid'**: Customer claims they already sent the money. The bot notes this down for human review.
- **'Wrong-Person'**: Caller says they are not Rahul Sharma. The bot apologizes and ends the call.
- **Data Extracted**: The bot automatically saves the 'Promise-to-Pay Date' (PTP Date) and 'Payment Amount'.

4. AUTOMATED TOOLS & ACTIONS TABLE

Function Name	Inputs	Outputs	Description
`verify_customer`	`input_dob` (string)	`{"auth_status": "success"}`	Checks if the spoken Date of Birth matches the database records.
log_promise_to_pay`	`ptp_date` (string)	`{"status": "logged"}`	Saves the customer's promised payment date directly into the company CRM.
send_payment_link`	`phone_number` (str)	`{"status": "sms_sent"}`	Instantly dispatches a secure payment link via SMS/WhatsApp to the customer.
escalate_to_agent	None	`{"status": "transferring"}`	Instantly transfers the live telephone call to a human representative.

5. LEGAL SAFETY & COMPLIANCE GUARDRAILS

- **Third-Party Privacy:** If a spouse, child, or colleague answers the phone, the bot will never reveal that the call is about an overdue loan. It will simply ask for a callback.
- **Legal Calling Hours:** Outbound calls are programmatically restricted to lawful hours only (8:00 AM to 7:00 PM).
- **Behavioral Guardrails:** The bot uses a polite, professional tone. Threats, high-pressure harassment, or rude behavior are completely blocked.
- **Hallucination Control:** If the user tries to distract the bot (e.g., asking "What's the weather today?"), the bot politely redirects them back to the account discussion.

6. HANDLING DIFFICULT SCENARIOS (EDGE CASES)

- **Abusive Callers:** If the customer uses profanity, the bot issues one polite warning. If they swear again, it logs the call as 'Hostile' and transfers or ends the call.
- **Language Switching:** If the customer switches from English to Hindi or Hinglish mid-call (e.g., "अभी पैसे नहीं हैं, कल दूँगा"), the bot switches its language instantly without losing track of the conversation steps.
- **Logged Call Outcomes (Dispositions):** Every call is automatically tagged in the database with a final outcome label: 'PTP_Collected', 'Left_Voicemail', 'Customer_Dispute', 'Verification_Failed', or 'Hostile_Terminated'.

7. MONITORING AND IMPROVING THE BOT (OBSERVABILITY FORMULAS)

To ensure the bot is performing well, the operations team tracks four primary metrics using these mathematical calculations:

- **Containment Rate:** Measures the % of calls handled completely by the bot without human intervention. Formula: $(\text{Total Calls Resolved by Bot} / \text{Total Answered Calls}) \times 100$
- **PTP (Promise to Pay) Rate:** Measures successful commitments collected. Formula: $(\text{Total Confirmed Payment Commitments} / \text{Total Successfully Authenticated Calls}) \times 100$

- **Average Latency:** The total round-trip time it takes for the bot to respond after a human stops talking. Target: Under 1.2 seconds total across STT, LLM, and TTS generation cycles.
- **Drop Rate:** The percentage of callers who hang up mid-conversation. Formula: $(\text{Total Calls with Mid-Call User Hang-Up} / \text{Total Connected Calls}) \times 100$

8. INFRASTRUCTURE ARCHITECTURE DIAGRAM

